



ANSWER KEY

1. (a) 2. (a) 3. (d) 4. (d) 5. (d) 6. (b) 7. (c) 8. (c) 9. (c) 10. (a)
 11. (d) 12. (c) 13. (d) 14. (c) 15. (c) 16. (a) 17. (c) 18. (c) 19. [132] 20. [40]
 21. [32] 22. [0.75] 23. [-245] 24. [4] 25. [375] 26. [120] 27. [10] 28. [1] 29. [2] 30. [16]

Hints and Solutions

1. (a)
 2. (a) Work done = area under F vs x diagram.
 $\therefore W_3 > W_2 > W_1 > W_4$
 3. (d) $W = \int \vec{F} \cdot d\vec{s}$
 $= (-x\hat{i} + y\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$
 $= \int_1^0 -x dx + \int_0^1 y dy$
 $= \left. \frac{x^2}{2} \right|_1^0 + \left. \frac{y^2}{2} \right|_0^1 = \left(0 + \frac{1}{2} \right) + \left(\frac{1}{2} \right) = 1\text{J}$
 4. (d) $W = F s \cos \theta = m \left(g + \frac{g}{2} \right) \times \left[\frac{1}{2} \times \frac{g}{2} \times t^2 \right] = \frac{3}{8} mg^2 t^2$
 5. (d) By work energy theorem work done by net force = $\Delta \text{K.E.}$

$$\Rightarrow W = \frac{1}{2} mv_f^2 - \frac{1}{2} mv_i^2$$

$$W = \frac{1}{2} \times 0.5 \times (0.25)^2 \times (4)^5$$

$$W = 16\text{ J}$$

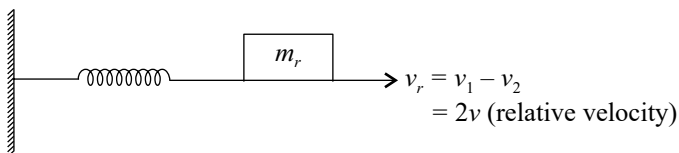
6. (b) Let $m = 250\text{g} = 0.25\text{kg}$



Reduced mass of the system

$$m_r = \frac{m_1 m_2}{m_1 + m_2} = \frac{mm}{m+m} = \frac{m}{2}$$

From conservation of energy,



$$K.E_i + P.E_i = K.E_f + P.E_f$$

$$\frac{1}{2} m_r v_r^2 + 0 = 0 + \frac{1}{2} kx^2$$

$$= \frac{1}{2} \times \frac{m}{2} \times 4v^2 = \frac{1}{2} kx^2$$

$$\frac{kx^2}{2} = mv^2$$

$$\frac{2}{2} x^2 = 0.25v^2$$

$$x = 0.5v$$

$$x = \frac{v}{2}$$

7. (c) Work done by retarding force = Change in kinetic energy

$$\Rightarrow \int_{x_i}^{x_f} -kx dx = \frac{1}{2} m [V_f^2 - V_i^2]$$

$$\Rightarrow \frac{-k}{2} [x_f^2 - x_i^2]_{0.5}^{1.5} = \frac{1}{2} \times 2 [V_f^2 - 16] \quad [\because V_i = 4\text{ m/s}]$$

$$\Rightarrow \frac{-12}{2} [1.5^2 - 0.5^2] = [V_f^2 - 16]$$

$$\Rightarrow -6[2 \times 1] = V_f^2 - 16$$

$$\Rightarrow v_f^2 = 16 - 12 = 4$$

$$\therefore v_f = 2\text{ m/sec.}$$

8. (c) $\vec{F} = 4x\hat{i} + 3y^2\hat{j}$

$$d\vec{r} = d\vec{s} = dx\hat{i} + dy\hat{j}$$

$$W = \int \vec{F} \cdot d\vec{r}$$

$$= \int (4x\hat{i} + 3y^2\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_1^2 4x dx + \int_2^3 3y^2 dy = \frac{4}{2} [x^2]_1^2 + \frac{3}{3} [y^3]_2^3$$

$$= 2[4 - 1] + [27 - 8] = 6 + 19 = 25 \text{ J}$$

From work-energy theorem

$$\Delta K = W = 25 \text{ J}$$

9. (c) According to conservation of energy, we can write

Loss in potential energy = Gain in kinetic energy

$$U_{in} - U_f = K_f - K_{in}$$

$$mg y - mg y_0 = K_f - 0$$

$$K_f = mg(y - y_0)$$

10. (a) Work done = Change in K.E.

$$\text{So } W_{mg} + W_2 = \frac{1}{2} M (0.8\sqrt{gh})^2 - 0$$

[W_2 work done by air friction]

$$\Rightarrow W_2 = 0.32mgh - mgh = -0.68mgh$$

$$\therefore W_2 = -0.68mgh$$

11. (d) $K.E. - 3 = \vec{F} \cdot \vec{d}$

$$K.E. = 3 + (3\hat{i} - 12\hat{j}) \times (4\hat{i})$$

$$K.E. = 3 + 12 = 15 \text{ J}$$

12. (c) Using linear conservation of momentum:

$$mv = (m + 4m)v_f \quad \dots(i)$$

Using conservation of energy,

$$\frac{1}{2}mv^2 = \frac{1}{2}(m + 4m)v_f^2 + mgh \quad \dots(ii)$$

Using equation (i) and (ii)

$$h = \frac{2v^2}{5g}$$

13. (d) When $v_{\max} \Rightarrow$ acceleration = 0

$$\Rightarrow x = \frac{F}{k}$$

Applying work energy theorem

$$W_{sp} + W_F = \Delta K.E.$$

$$-\frac{1}{2}kx^2 + F.x = \Delta K.E.; -\frac{1}{2}k\frac{F^2}{k^2} + \frac{F^2}{k} = \frac{1}{2}mv_{\max}^2$$

$$\frac{F^2}{2k} = \frac{1}{2}mv_{\max}^2 \Rightarrow \frac{F}{\sqrt{mk}} = v_{\max}$$

14. (c) Centripetal acceleration, $a = k^2 r t^2$

Since, centripetal acceleration is $\frac{v^2}{r}$

$$\therefore \frac{v^2}{r} = k^2 r t^2$$

$$v = k r t$$

Centripetal force is perpendicular to the tangential velocity, therefore power delivered by it will be zero.

Tangential acceleration,

$$\Rightarrow a_t = \frac{dv}{dt} = kr$$

$$\Rightarrow \text{Tangential force, } F_t = ma_t = mkr$$

$$\therefore \text{Power delivered, } P = F_t v = (mkr)(krt) = mk^2 r^2 t$$

$$15. (c) \int_0^v v dv = \frac{P}{m} \int_0^t dt \Rightarrow v = \left(\sqrt{\frac{2P}{m}} \right) t^{1/2} \Rightarrow v \propto t^{1/2}$$

$$\Rightarrow x \propto t^{3/2} \left(\because v = \frac{x}{t} \right)$$

16. (a) Power, $P = \text{constant}$

$$P = Fv = mv \frac{dv}{dx} v$$

$$\frac{P}{m} \int dx = \int v^2 dv$$

$$\frac{Px}{m} = \frac{v^3}{3}$$

$$\left(\frac{3P}{m} x \right)^{1/3} = v = \frac{dx}{dt}$$

$$\left(\frac{3P}{m} \right)^{1/3} \int_0^t dt = \int_0^x x^{-1/3} dx \Rightarrow x = \left(\frac{8P}{9m} \right)^{1/2} t^{3/2}$$

$$17. (c) \text{At equilibrium, } \frac{dU}{dr} = 0 \Rightarrow \frac{d}{dr} \left[\frac{A}{r^{10}} \right] - \frac{d}{dr} \left[\frac{B}{r^5} \right] = 0$$

$$\Rightarrow -10Ar^{-10-1} - B(-5)r^{-5-1} = 0$$

$$\Rightarrow -10Ar^{-11} + 5Br^{-6} = 0$$

$$10Ar^{-11} = 5Br^{-6}$$

$$\Rightarrow r^5 = \frac{2A}{B}$$

$$\therefore r = \left(\frac{2A}{B} \right)^{1/5}$$

18. (c) $ME = PE + KE = 8J$

$$\text{At } x_3, 8 = KE + 4 \Rightarrow KE = 4J$$

$$\text{At } x_2, 8 = KE + 0 \Rightarrow KE_{\max} = 8J$$

$$\text{At } x < x_1, 8 = KE + 8 \Rightarrow KE \text{ is } 0$$

19. [132] $\because F = 5 + 3y^2$

$$\Rightarrow W = \int_2^5 (5 + 3y^2) dy$$

$$= \left[5y + \frac{3y^3}{3} \right]_2^5$$

$$\therefore W = 132 \text{ J}$$

20. [40] $\because W = \vec{F} \cdot (\vec{r}_f - \vec{r}_i)$

$$W = (5\hat{i} + 2\hat{j} + 7\hat{k}) \cdot ((5\hat{i} - 2\hat{j} + \hat{k}) - (2\hat{i} + 3\hat{j} - 4\hat{k}))$$

$$W = 40 \text{ J}$$

21. [32] Formula of work done $W = \int_{x_1}^{x_2} F(x) \cdot dx$

$$W = \int_0^4 (2 + 3x) dx = \left[2x + \frac{3x^2}{2} \right]_0^4$$

$$= 8 + 3 \times 8$$

$$= 32 \text{ J}$$

22. [0.75]

$$\vec{F} = (\alpha y \hat{i} + 2\alpha x \hat{j})$$

$$W_{AB} = (-1\hat{i}) \cdot (1\hat{i}) = -1 \text{ J}$$

$$\begin{bmatrix} \vec{F} = -1\hat{i} + 2\alpha x \hat{j} \\ \vec{S} = 1\hat{i} \end{bmatrix}$$

Similarly,

$$W_{BC} = 1 \text{ J}$$

$$W_{CD} = 0.25 \text{ J}$$

$$W_{DE} = 0.5 \text{ J}$$

$$W_{EF} = W_{FA} = 0.75 \text{ J}$$

$$\therefore \text{Net work in cycle} = 0$$

23. [-245] Using work energy theorem,

$$W_f + W_g = \Delta KE$$

$$W_f + 10 \times 0.3 = -\frac{1}{2} \times 1 \times (22)^2$$

$$W_f = -245 \text{ J}$$

24. [4] $\frac{1}{2}mv^2 = Pt$

$$v = \sqrt{\frac{2Pt}{m}}$$

$$\frac{dx}{dt} = \sqrt{\frac{2Pt}{m}}$$

$$dx = \sqrt{\frac{2Pt}{m}} dt$$

Integration both sides

$$x = \sqrt{\frac{P}{m}} \frac{2}{3} [t^{3/2}]_0^4$$

$$x = \frac{16\sqrt{P}}{3} = \frac{1}{3} \times 16\sqrt{P}$$

$$\text{Also, } x = \frac{1}{3} \alpha^2 \sqrt{P}$$

$$\Rightarrow \alpha = 4$$

25. [375] Let v_1 and v_2 are velocity just before and just after hitting the floor.

$$\frac{v_1}{v_2} = 4 \Rightarrow v_2 = \frac{v_1}{4}$$

$$KE_1 = \frac{1}{2}mv_1^2$$

$$KE_2 = \frac{1}{2}mv_2^2 = \frac{1}{2}m \cdot \frac{v_1^2}{16}$$

$$\Delta KE = \frac{1}{2}mv_1^2 \left(\frac{1}{16} - 1 \right) = \frac{-15}{32}mv_1^2$$

$$\% \text{ loss in kinetic energy} = \frac{\Delta KE}{KE_1} \times 100\%$$

$$= \frac{-15}{16} \times 100 = \frac{-375}{4} \%$$

26. [120] Conservation of energy $K.E_1 + P.E_1 = K.E_2 + P.E_2$

$$\Rightarrow 0 + mg \left(h + \frac{h}{2} \right) = 0 + \frac{1}{2}k \left(\frac{h}{2} \right)^2$$

$$\Rightarrow 3mgh = \frac{k}{4}h^2 \Rightarrow k = \frac{12mg}{h} \Rightarrow k = \frac{12 \times 0.1 \times 10}{0.1}$$

$$k = 120 \text{ Nm}^{-1}$$

27. [10] By energy conservation

$$K_i + U_i = K_f + U_f$$

$$0 + 10 \times 10 \times 10 = \frac{1}{2} \times 10 \times v_B^2 + 10 \times 10 \times 5$$

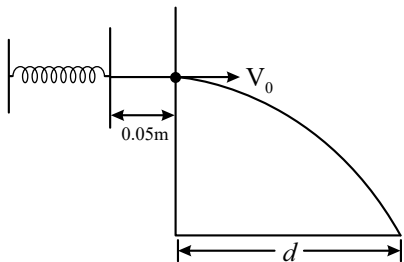
$$1000 = 5v_B^2 + 500$$

$$v_B^2 = \frac{500}{5} = 100$$

$$v_B = 10 \text{ m/s}$$

$$x = 10$$

28. [1] Let ball starts its motion with horizontal velocity v_0 , so with the help of conservation of mechanical energy, we can write



$$\frac{1}{2}mv_0^2 = \frac{1}{2}kx^2 \Rightarrow v_0 = x\sqrt{\frac{k}{m}} = 0.05 \times \sqrt{\frac{100}{0.1}}$$

$$= 0.5 \times \sqrt{10} \text{ m/s}$$

For vertical motion,

$$t = \text{Time required to fall the ball} = \sqrt{\frac{2h}{g}} = 2 \times \sqrt{\frac{1}{10}} \text{ s}$$

For horizontal motion,

$$\Rightarrow d = v_0 \times t = 0.5 \times \sqrt{10} \times 2 \times \sqrt{\frac{1}{10}} = 1 \text{ m}$$

29. [2] $mg \cos \theta = \frac{mv^2}{R} \quad \dots (i)$

$\cos \theta = \frac{h}{R} \quad \dots (ii)$

Loss in potential Energy = Gain in kinetic Energy

$$\Rightarrow mg(R-h) = \frac{1}{2}mv^2 \quad \dots (iii)$$

Using (i), (ii) and (iii)

$$\Rightarrow mg\left(\frac{h}{R}\right) = \frac{2mg(R-h)}{R}$$

$$\Rightarrow h = \frac{2R}{3} = 2 \text{ m}$$

Hence, value of h is 2m

30. [16] Work done by friction + work done by spring

$$= 0 - \frac{1}{2}mV^2 \quad [\text{As 90\% of K.E. is lost by friction}]$$

$$-\frac{90}{100}\left(\frac{1}{2}mV^2\right) - \frac{1}{2}Kx^2 = -\frac{1}{2}mV^2$$

$$-\frac{1}{2}Kx^2 = \left(\frac{90}{100} - 1\right)\frac{1}{2}mV^2 = \frac{-10}{100}\frac{1}{2}mV^2$$

$$-Kx^2 = -\frac{1}{10}mV^2$$

$$-K = -\frac{1}{10} \times \frac{40000 \times 20^2}{1} = 4000 \times 400$$

$$-K = -16 \times 10^5$$

$$K = 16 \times 10^5$$

